EduDropout Predictor: Early Identification of Students at Risk of Dropping Out

Introduction

Education plays a crucial role in shaping the future of individuals and society. However, student dropout remains a persistent challenge across many educational systems worldwide. A student dropout refers to a situation where a student leaves school or educational programs before completing their studies. This issue is influenced by various academic, personal, and socio-economic factors such as poor academic performance, lack of motivation, financial constraints, and insufficient support systems.

In recent years, the integration of data science and machine learning techniques has provided new opportunities to address such problems effectively. By analyzing historical student data, it is possible to identify patterns and trends that indicate whether a student is at risk of dropping out. Early identification allows educational institutions to take preventive measures such as counseling, mentoring, or academic support.

This project aims to develop a predictive system that can identify students at risk of dropping out using machine learning techniques. The model is built using student academic records and behavioral data to classify students into at-risk and not-at-risk categories. The project contributes to improving educational outcomes and aligns with global goals such as ensuring quality education for all.

Problem Statement

Student dropout is a significant issue that negatively impacts both individuals and society. When students drop out of school, they often face limited career opportunities, reduced earning potential, and social challenges. Educational institutions also suffer due to reduced performance metrics and inefficiencies in resource utilization.

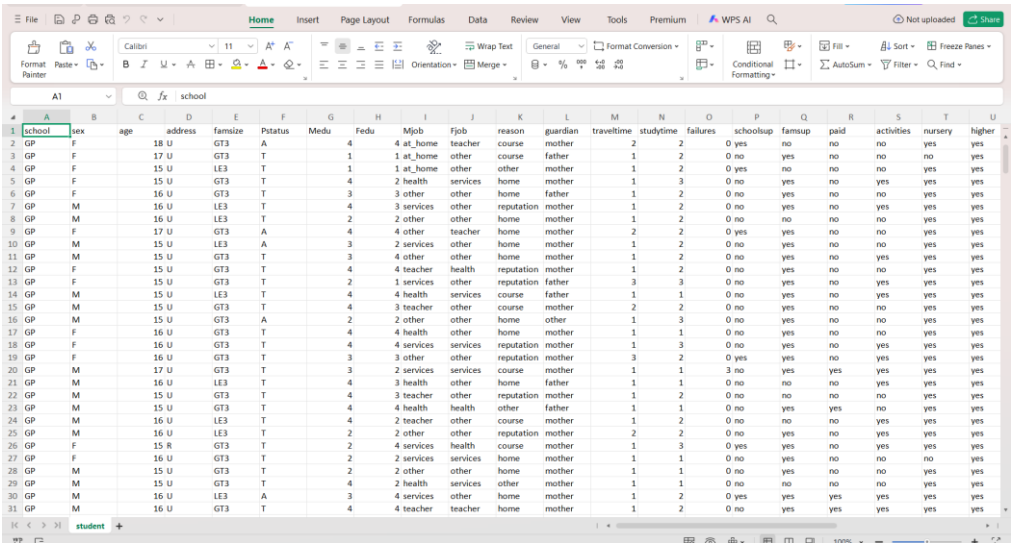
Traditional approaches to identifying at-risk students are often reactive and based on manual observation by teachers or administrators. These methods may fail to detect early warning signs, leading to delayed interventions. As a result, many students who could have been supported end up leaving the education system.

The main objective of this project is to develop a data-driven solution that can predict student dropout risk at an early stage. By using machine learning algorithms, the system can analyze multiple factors simultaneously and provide accurate predictions. This enables educators to take timely actions and implement targeted interventions to support students.

The problem can be framed as a binary classification task, where each student is classified as either at risk of dropping out or not. The success of the model depends on the quality of data and the effectiveness of feature selection and preprocessing techniques.

Dataset Description

The dataset used in this project is the Student Performance Dataset, which is widely used for educational data analysis and machine learning applications. It contains detailed information about students' academic performance along with various personal, social, and school-related attributes. This dataset helps in understanding the factors that influence student success and dropout risk and is particularly suitable for classification problems such as predicting whether a student is at risk of dropping out. The dataset was obtained from Kaggle, a popular platform for data science and machine learning datasets, and the file used is *student-por.csv*, belonging to the domain of Education Analytics. It consists of approximately 649 student records and more than 30 attributes, making it a structured tabular dataset where each row represents a student and each column represents a specific feature.



school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	reason	guardian	traveltime	studytime	failures	schoolsup	famsup	paid	activities	nursery	higher
GP	F	18	U	GT3	A	4	4	at_home	teacher	course	mother	2	2	0	yes	no	no	no	yes	yes
GP	F	17	U	GT3	T	1	1	at_home	other	course	father	1	2	0	no	yes	no	no	no	yes
GP	F	15	U	LE3	T	1	1	at_home	other	other	mother	1	2	0	yes	no	no	no	yes	yes
GP	F	15	U	GT3	T	4	2	health	services	home	mother	1	3	0	no	yes	no	yes	yes	yes
GP	F	16	U	GT3	T	3	3	other	other	home	father	1	2	0	no	yes	no	no	yes	yes
GP	M	16	U	LE3	T	4	3	services	other	reputation	mother	1	2	0	no	yes	no	yes	yes	yes
GP	M	16	U	LE3	T	2	2	other	other	home	mother	1	2	0	no	no	no	no	yes	yes
GP	F	17	U	GT3	A	4	4	other	teacher	home	mother	2	2	0	yes	yes	no	no	yes	yes
GP	M	15	U	LE3	A	3	2	services	other	home	mother	1	2	0	no	yes	no	no	yes	yes
GP	M	15	U	GT3	T	3	4	other	other	home	mother	1	2	0	no	yes	no	yes	yes	yes
GP	F	15	U	GT3	T	4	4	teacher	health	reputation	mother	1	2	0	no	yes	no	no	yes	yes
GP	F	15	U	GT3	T	2	1	services	other	reputation	father	3	3	0	no	yes	no	yes	yes	yes
GP	M	15	U	LE3	T	4	4	health	services	course	father	1	1	0	no	yes	no	yes	yes	yes
GP	M	15	U	GT3	T	4	3	teacher	other	course	mother	2	2	0	no	yes	no	no	yes	yes
GP	M	15	U	GT3	A	2	2	other	other	home	other	1	3	0	no	yes	no	no	yes	yes
GP	F	16	U	GT3	T	4	4	health	other	home	mother	1	1	0	no	yes	no	no	yes	yes
GP	F	16	U	GT3	T	4	4	services	services	reputation	mother	1	3	0	no	yes	no	yes	yes	yes
GP	F	16	U	GT3	T	3	3	other	other	reputation	mother	3	2	0	yes	yes	no	yes	yes	yes
GP	M	17	U	GT3	T	3	2	services	services	course	mother	1	1	3	no	yes	yes	yes	yes	yes
GP	M	16	U	LE3	T	4	3	health	other	home	father	1	1	0	no	no	no	yes	yes	yes
GP	M	15	U	GT3	T	4	3	teacher	other	reputation	mother	1	2	0	no	no	no	no	yes	yes
GP	M	15	U	GT3	T	4	4	health	health	other	father	1	1	0	no	yes	yes	no	yes	yes
GP	M	16	U	LE3	T	4	2	teacher	other	course	mother	1	2	0	no	no	no	yes	yes	yes
GP	M	16	U	LE3	T	2	2	other	other	reputation	mother	2	2	0	no	yes	no	yes	yes	yes
GP	F	15	R	GT3	T	2	4	services	health	course	mother	1	3	0	yes	yes	no	yes	yes	yes
GP	F	16	U	GT3	T	2	2	services	services	home	mother	1	1	0	no	yes	no	no	no	yes
GP	M	15	U	GT3	T	2	2	other	other	home	mother	1	1	0	no	yes	no	no	yes	yes
GP	M	15	U	GT3	T	4	2	health	services	other	mother	1	1	0	no	no	no	no	yes	yes
GP	M	16	U	LE3	A	3	4	services	other	home	mother	1	2	0	yes	yes	yes	yes	yes	yes
GP	M	16	U	GT3	T	4	4	teacher	teacher	home	mother	1	2	0	no	yes	yes	yes	yes	yes

The features in the dataset can be broadly categorized into academic, behavioral, and socio-economic factors. Academic features include grades (G1, G2, G3), study time, and the number of past failures, which directly indicate student performance. Behavioral features include absences, free time, social activities, and participation in extracurricular activities, reflecting student habits and engagement levels. Socio-

economic features include variables such as school, gender, age, address (urban or rural), family size, parental status, parental education levels, and educational support, all of which influence the learning environment and academic outcomes of students.

Since the dataset does not include a direct dropout label, a new target variable called “dropout” was created. The logic used is based on the final grade (G3), where students scoring less than 10 are considered at risk of dropping out (assigned a value of 1), and those scoring 10 or above are considered not at risk (assigned a value of 0). This transformation converts the dataset into a binary classification problem, which is suitable for machine learning models such as Logistic Regression.

The dataset contains both categorical and numerical variables and requires preprocessing steps such as encoding categorical features into numerical form. It does not contain major missing values, making it clean and suitable for analysis. Overall, this dataset is important because it reflects real-world student behavior, includes multiple influencing factors, helps identify early warning signs of dropout, and supports data-driven decision-making in education systems.

Data Preprocessing

Data preprocessing is a crucial step in any data science project, as raw data often contains inconsistencies, missing values, and irrelevant features. Proper preprocessing ensures that the data is clean, structured, and suitable for machine learning algorithms.

In this project, several preprocessing steps were performed. First, unnecessary columns such as intermediate grades (G1 and G2) were removed to avoid redundancy, as the final grade (G3) is more relevant for determining dropout risk. Next, categorical variables such as gender, school type, and family background were converted into numerical format using one-hot encoding. This step is necessary because machine learning models require numerical input.

```python
# Show raw data (before preprocessing)
df_raw = pd.read_csv("../dataset/raw_data/student.csv")
df_raw.head()
```

[3] ✓ 0.0s Python

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	absences	G
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	4	0
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	2	9
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	6	12
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	0	14
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	0	11

5 rows × 33 columns

Before cleaning the Dataset

```python
df_clean = pd.read_csv("../dataset/processed_data/cleaned.csv")
df_clean.head()
```

[4] ✓ 0.0s Python

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout	Dalc	...	guardian_mother	guardian_other	schoolsup_yes	famsu
0	18	4	4	2	2	0	4	3	4	1	...	True	False	True	
1	17	1	1	1	2	0	5	3	3	1	...	False	False	False	
2	15	1	1	1	2	0	4	3	2	2	...	True	False	True	
3	15	4	2	1	3	0	3	2	2	1	...	True	False	False	
4	16	3	3	1	2	0	4	3	2	1	...	False	False	False	

5 rows × 41 columns

After cleaning the Dataset

A new target variable called “dropout” was created based on the final grade. This variable indicates whether a student is at risk of dropping out. Additionally, the dataset was checked for missing values and inconsistencies, and appropriate cleaning techniques were applied.

These preprocessing steps significantly improve the quality of the dataset and enhance the performance of the machine learning model. Without proper preprocessing, the model may produce inaccurate or misleading results.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is used to understand the dataset and identify patterns, trends, and relationships among variables. It helps in gaining insights before building the predictive model.

In this project, EDA revealed several important observations. The distribution of the target variable showed that most students are not at risk of dropping out, while a smaller portion falls into the at-risk category. This indicates that the dataset is slightly imbalanced.

Further analysis showed that students with lower study time tend to have poorer academic performance, increasing their likelihood of dropping out. Similarly, students with higher absences were more likely to be at risk, as frequent absence negatively affects learning.

EDA also helped in identifying correlations between features. For example, parental education and family support showed some influence on student performance. Understanding these relationships helps in selecting relevant features for the model.

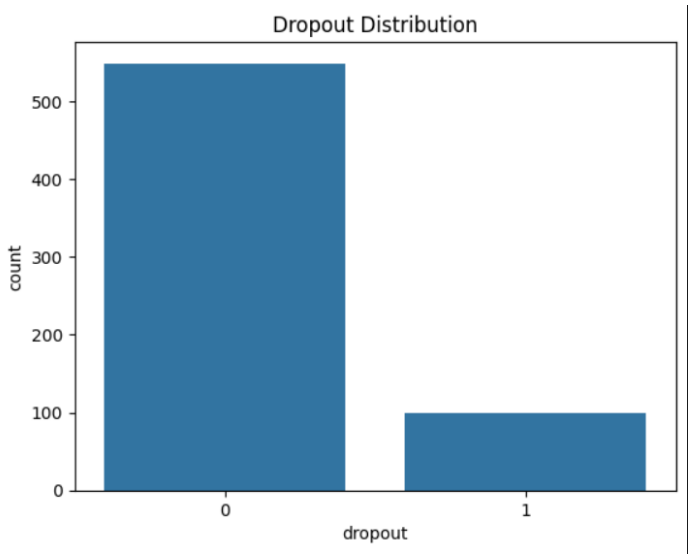
Overall, EDA provides valuable insights that guide the modeling process and improve prediction accuracy.

Visualization

Data visualization is an important step in data analysis as it helps in representing complex data in a simple and understandable graphical format. In this project, various visualization techniques were used to analyze the relationship between different features and student dropout risk. These visualizations provide clear insights into patterns and trends that are not easily observable from raw data.

1. Dropout Distribution Visualization

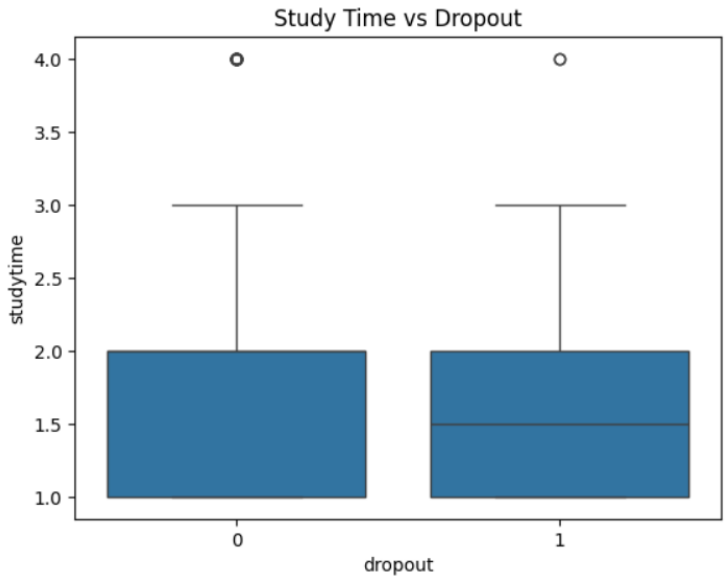
Figure 1: Dropout Distribution



The dropout distribution graph represents the number of students who are at risk and not at risk of dropping out. It is observed that the majority of students fall under the “not at risk” category, while a smaller portion belongs to the “at risk” category. This indicates that the dataset is slightly imbalanced. Such imbalance is important to consider when building machine learning models, as it may affect prediction accuracy.

2. Study Time vs Dropout Visualization

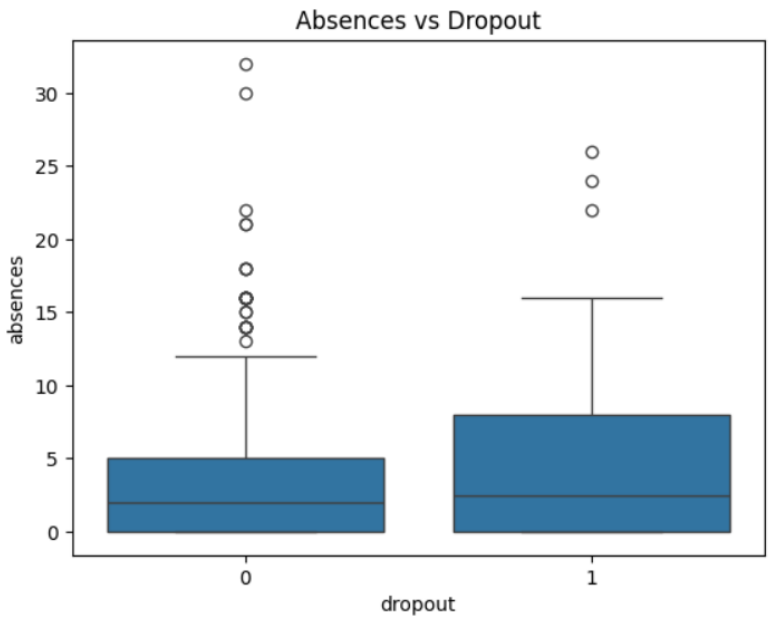
Figure 2: Study Time vs Dropout



This visualization shows the relationship between study time and dropout risk. It can be observed that students with lower study time tend to have a higher probability of dropping out. Students who dedicate more time to studying generally perform better academically and are less likely to be at risk. This highlights the importance of consistent study habits.

3. Absences vs Dropout Visualization

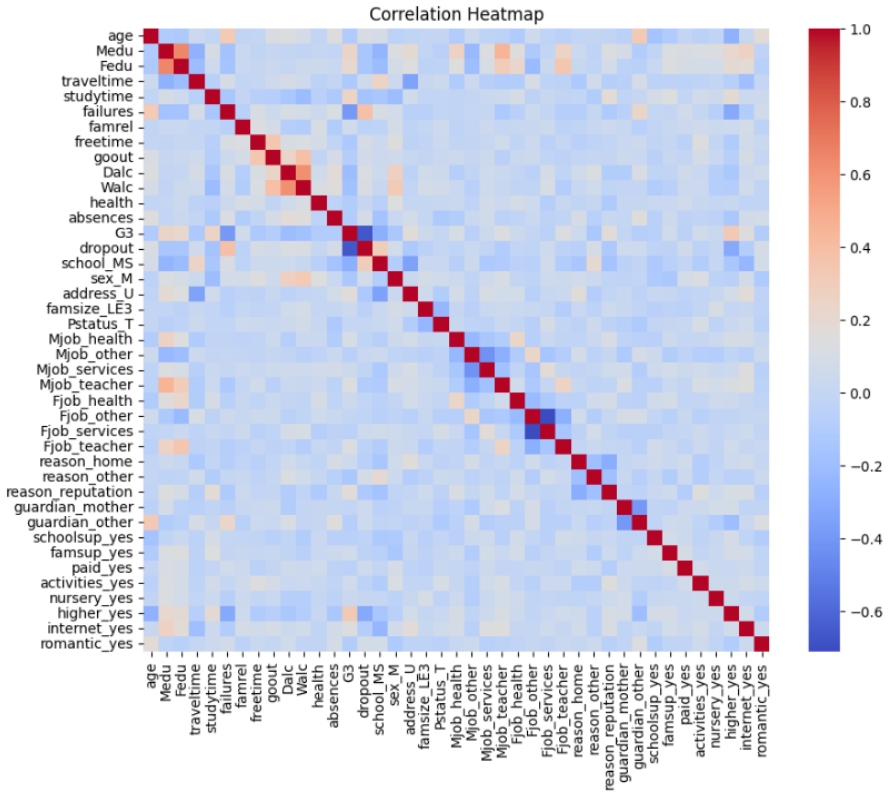
Figure 3: Absences vs Dropout



The absences versus dropout graph shows that students with a higher number of absences are more likely to be at risk of dropping out. Frequent absence from school leads to missed lessons and reduced academic performance, which increases dropout risk. This visualization emphasizes the importance of regular attendance.

4. Correlation Heatmap Visualization

Figure 4: Correlation Heatmap



The correlation heatmap shows the relationship between different features in the dataset. It helps in identifying which variables are strongly related to each other. Academic features such as grades show strong correlations, while behavioral and socio-economic factors show moderate relationships. This visualization is useful for feature selection and understanding the overall structure of the dataset.

Model Building

Model building is the process of selecting, training, and evaluating a machine learning algorithm to make predictions based on the given dataset. In this project, Logistic Regression was used as the primary model to predict whether a student is at risk of dropping out. Logistic Regression is a supervised learning algorithm commonly used for binary classification problems. It works by estimating the probability that a given input belongs to a particular class using a sigmoid function, which outputs values between 0 and 1. Based on a predefined threshold (usually 0.5), the model classifies students into two categories: “at risk” and “not at risk.” The simplicity, efficiency, and interpretability of Logistic Regression make it a suitable choice for this problem.

Before training the model, the dataset was divided into two parts: the training set and the testing set. Typically, 80% of the data is used for training and 20% for testing. The input features (X), which include variables such as study time, absences, and socio-economic factors, were separated from the target variable (y), which represents the dropout status. The training dataset is used to teach the model by allowing it to learn the relationships between the input features and the target variable. During this phase, the model adjusts its internal parameters to minimize prediction errors and capture meaningful patterns in the data.

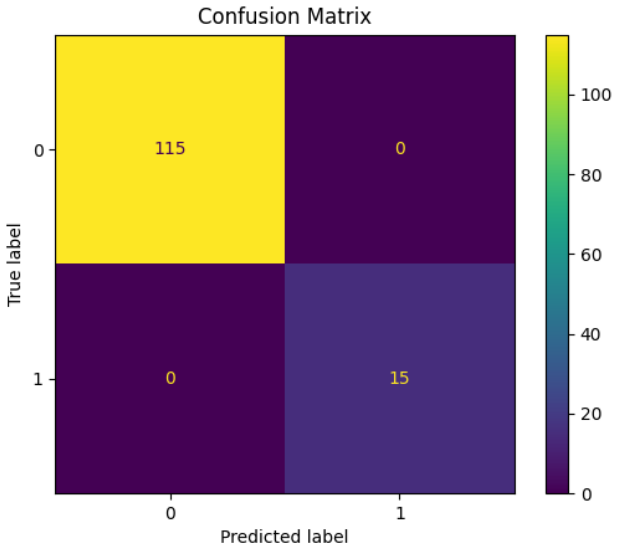
Once the model is trained, it is applied to the testing dataset to make predictions on unseen data. This step is important to evaluate how well the model generalizes beyond the training data. The predicted values are then compared with the actual values to assess the model’s performance. This ensures that the model is not simply memorizing the training data but is capable of making accurate predictions on new data.

To evaluate the effectiveness of the model, several performance metrics were used, including accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of the model, while precision and recall provide insights into how well the model identifies students at risk.

Results

The model achieved high accuracy in predicting student dropout risk. Evaluation metrics such as accuracy, precision, recall, and F1-score were used to assess performance.

The confusion matrix shows that the model correctly classified most students, with very few misclassifications. This indicates that the model is reliable and effective.



The classification report provides detailed insights into the performance for each class. High precision and recall values indicate that the model performs well in identifying both at-risk and non-risk students.

Although the model achieved very high accuracy, it is important to note that the dataset is relatively simple, which may lead to overfitting. In real-world scenarios, performance may vary.

Conclusion

This project demonstrates the effectiveness of machine learning in addressing real-world problems such as student dropout. By analyzing academic and socio-economic data, it is possible to identify students at risk and take preventive actions.

The use of Logistic Regression provided a simple yet powerful solution for classification. The results show that factors such as study time, attendance, and academic performance play a significant role in determining dropout risk.

The project highlights the importance of data-driven decision-making in education. By implementing such systems, institutions can improve student retention and overall performance.

Future improvements can include using advanced models, incorporating more data, and deploying the system as a real-time application. Overall, this project provides a strong foundation for further research and development in education analytics.